AA 273: State Estimation and Filtering for Robot Perception

Instructor
Prof. Mac Schwager
Office: Durand 266
email: schwager@stanford.edu
Office hours: Tues, 3-4pm

Course Assistant
Lachlain McGranahan
email: lachmcg@stanford.edu
Office hours: Mondays, 5-6pm, Durand 270
Thursdays, 3-4pm, Durand 270

Course Assistant
Sidharth Anantha
email: ananthas@stanford.edu
Office hours: Fridays, 10-11am, Durand 270
Thursdays, 3-4pm, Durand 270

Meeting Time and Place

Time: Tues and Thurs, 1:30pm–2:50pm
Place: Thornton 102

Description

How does an autonomous aircraft determine its velocity vector and position with noisy GPS, airspeed, and IMU measurements? How does an autonomous car map its surrounding environment and determine its position relative to that environment with noisy camera and Lidar measurements? How does a satellite determine its position and orientation relative to the Earth using noisy inertial sensors? These are all examples of state estimation, the problem of determining the state of a dynamical system given noisy measurements over time. This course will cover Kalman filtering, recursive Bayesian filtering, and nonlinear filter architectures including the extended Kalman filter, particle filter, and unscented Kalman filter. The course will stress examples from robotics, including state estimation for drones, SLAM for autonomous cars and mobile robots, and attitude estimation for autonomous spacecraft. We will also seek to put state estimation into a larger context of Bayesian estimation relevant to a range of topics in machine learning, artificial intelligence, and signal processing.

Prerequisites

The course will assume knowledge of concepts from state space control and linear systems (ENGR 205, AA212, or EE 263), and basic familiarity with the Kalman filter. A high level of mathematical maturity will be expected from the students.

Structure and Expectations

The course will have four main components: lectures, problem sets, a midterm, and a final project.

Lectures Students will be expected to attend lectures. Video recordings of the lectures will be available on the Canvas website.

Problem Sets Problem sets will be assigned regularly throughout the quarter. They will include a mix of traditional mathematical problems and implementation exercises in Matlab or Python. Of course, all solutions should contain your own original work. You can turn in your problem sets through the course gradescope page.

Midterm We will have a take-home midterm covering the material in the first half of the class. The midterm will be a 5 hour timed exam, which you can choose to take within a 24 hour window. More details will follow in class.

Project There will be a significant group project component to this class involving two steps. You will form groups of up to four students. First, groups will conduct a literature review in a relevant research area, and write a brief proposal for what they intend to do as a class project. Finally, groups will carry out the project, producing a written research report as well as an oral presentation for the class. Choosing a project that dovetails with your graduate research is highly encouraged.

Grading

All assignments should be submitted through the course gradescope page, and all grading will be done through gradescope. The final course grade will be assigned according to the following weighting:

Problem Sets		45%
Midterm		20%
Course project		35%
Project proposal	10%	
Final report & presentation	25%	

Course Websites

All course materials will be disseminated on the Canvas course website. A link to Canvas can be found at `https://canvas.stanford.edu/courses/`. We will also have an Ed page for class discussions, and a gradescope page for submitting and grading assignments.

Textbooks

Required:

T. D. Barfoot, *State Estimation for Robotics* Cambridge University Press, 2017. A pdf can be dowloaded for free at: `http://asrl.utias.utoronto.ca/~tdb/`.

S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*, MIT Press, 2006.

Optional:

D. Simon, *Optimal State Estimation: Kalman, H_∞, and Nonlinear Approaches*, Wiley and Sons, Inc., 2006.

Late Policy

We assume you will submit assignments on time. We will consider requests for late days if they are *requested by email at least 24 hours prior to the due date with a legitimate professional justification.* Traveling for a conference or your job, a serious illness, or a major research or professional deadline are legitimate excuses. Running out of time because it was a busy week is not. 10% per day will be deducted from assignments that are late without written approval from the teaching staff.

OAE Letters

If you have an Office of Accessible Education (OAE) letter, please submit it to the CAs at the beginning of the quarter. Please remind the CAs before the deadline if you plan to use OAE authorized late days for

an assignment. Similarly please remind the CAs before the exam if you use OAE authorized extra time on exams. This will minimize misgrading and will help the CAs to manage grading logistics.

AI Assistance

As specified by the Stanford Honor Code on Generative AI Policy:

Absent a clear statement from a course instructor, use of or consultation with generative AI shall be treated analogously to assistance from another person. In particular, using generative AI tools to substantially complete an assignment or exam (e.g. by entering exam or assignment questions) is not permitted. Students should acknowledge the use of generative AI (other than incidental use) and default to disclosing such assistance when in doubt.

In this class, we allow the use of AI coding assistants that are integrated into your IDE, Colab, Jupyter, or Matlab coding environment. We do not allow consultation with AI chatbots for PS or exam solutions or for writing proposal or project reports. You can use AI to help generate illustrative figures or tables for your reports, but do include a citation to clearly attribute AI usage for figures and tables. Graders will check for evidence of unauthorized AI usage. If in doubt, ask the teaching staff.

Planned Schedule

ProRo refers to the book Probabilistic Robotics, and SER refers to State Estimation for Robotics

Date	**Topic**	**Reading**	**Assignments**
1/6 T	Class Overview, Control Sys and Prob Review	ProRo 1	
1/8 Th	Rand Vars, Distributions, Bayes' Rule	ProRo 2, SER 2	PS 1 out
1/13 T	Rand Proc, Markov Proc, Graph Models	ProRo 2, SER 2	
1/15 Th	Bayesian Filter, multivariate Gaussian	ProRo 2, SER 2	PS 2 out, PS 1 due
1/20 T	Gaussian RVs, Moments, E[·], Cov(·)	ProRo 3.2, SER 2	
1/22 Th	Gaussian Estimation, Gaussian Processes	ProRo 3.2, SER 3.1	PS 3 out, PS 2 due
1/27 T	Linear Gaussian Systems	SER 3.3	
1/29 Th	Kalman Filter Derivation	ProRo 3.2, SER 3.1, 3.3	PS 4 out, PS 3 due
2/3 T	KF Properties, MMSE	ProRo 3.3, SER 3.3	
2/5 Th	Extended Kalman Filter (EKF, iEKF)	ProRo 3.4, SER 4.2	Midterm
2/10 T	Unscented Kalman Filter (UKF, iUKF)	ProRo 3.4, SER 4.2	
2/12 Th	Particle Filter (PF)	ProRo 4.3, SER 4.2	PS 5 out, PS 4 due
2/17 T	Non-Gaussian Est, MAP/ML Est	SER 4.3	Proposals due
2/19 Th	Rigid Body Poses, SLAM Front End/Back End	ProRo 6, SER 6.2, 6.3	PS 6 out, PS 5 due
3/24 T	Pose Graph Optimization Back End	ProRo 11, SER 8	
3/26 Th	Vision and Lidar Front Ends	ProRo 6, SER 6.4	
3/3 T	Camera Models, Rendering Equation		
3/5 Th	Special Topics: E.g. 3DGS, Diffusion Models		PS 6 due
3/10 Th	Project Presentations		
3/12 T	Project Presentations		Project reports due